VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY, NAGPUR

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PLACEMENT BROCHURE 2024-25

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About Us

The Department of Computer Science and Engineering, established in 1987, offers B.Tech., M. Tech. and Ph.D. degrees. The department has well qualified and motivated faculty members and support staff. The laboratories are adequately equipped with state-of-the-art facilities. The department is actively involved in R&D as well as consultancy projects and has collaborations with several industries, academic institutes and R&D organizations in the country. The first B. Tech batch graduated in 1991 and the first M.Tech. batch graduated in 2009. There are more than 20 full-time research scholars registered in the department.

Mission

To produce highly qualified and motivated graduates through a rigourous curriculum of theory and application that develops the ability to solve problems, individually and in teams. Creating knowledge of fundamental principles and innovative technologies through research within the core areas of computer science and also in inter-disciplinary topics. Serving the communities to which we belong at local and national levels, combined with a deep awareness of our ethical responsibilities to our profession and to society.

Vision

To contribute effectively to the important national endeavour to produce quality human resource in the information technology and related areas for sustainable development of the country's IT industry needs. To advance the state of the art in computer science and engineering by working on cutting edge research topics, publishing quality research papers and filing enduring patents. To serve the local and the national community by creating awareness about IT related products and to impress upon then the importance of knowledge management

From the HOD's desk

Dr. M. P. Kurhekar
Head of the Department,
Computer Science and Engineering.

VNIT Nagpur's Department of Computer Science & Engineering offers Bachelor of Technology, Master of Technology, and Doctor of Philosophy in Computer Science & Engineering. Our objective is to enhance human understanding of Computing through teaching and research, and to extend that understanding by solving real-world challenges. In teaching, our aim is to educate our students so that they have the fundamental and foundational knowledge as well as the interpersonal and technical skills needed to succeed and excel. In these efforts, students frequently collaborate with renowned research institutes and national and worldwide corporations, who open provide opportunities to work on their needs, which our students successfully fulfil.

VNIT's faculty and students at the Computer Science and Engineering department contribute to the computing sector by publishing patents and research work in peer-reviewed journals in many research fields, including Image Processing, Computer Vision, Artificial Intelligence, Machine Learning, Deep Learning, Cryptography, Computer and Network Security, Data Mining, Bioinformatics, Real-time Systems, large Cyber-Physical systems, and Parallel Computing.

Our students have brought enormous success to the department by demonstrating their programming skills in numerous programming competitions and hackathons.
The students are able to obtain prime jobs and secure admissions at prestigious national and international universities for higher education thanks to their technical expertise and interpersonal skills acquired during their stay at the VNIT campus.
I am confident that the students of the present batch will justify the credibility of the department by demonstrating a high level of professional competence in their respective future endeavours.

Dr. Mansi A. Radke

Faculty Coordinator, Training and Placement
Computer Science and Engineering.

Established in 1987, the Department of Computer Science and Engineering offers B. Tech., M. Tech., and Ph.D. degrees. The academics and support staff in the department are highly skilled and motivated. The laboratories are well equipped with cutting-edge technology and state-of-the-art facilities. The department is actively involved in R&D and consulting projects, and it collaborates with a number of industries, academic institutes, and R&D organisations throughout the country.

The department includes excellent laboratory facilities with an adequate number of desktop computers. In the laboratories, different kinds of servers, such as those manufactured by IBM, Apple, HP, and Dell, are installed. The latest Titan RTX GPU, which is the state-of-the-art GPU, was recently presented by the alumni batch of 1983 through the Alumni Association. Apart from that, we also have six other alumni donated servers which are being used in many of the latest projects in the area of machine learning by students and faculty for data-intensive tasks. Students from the department consistently perform well in national-level competitive examinations, programming contests, and hackathons such as the ACM ICPC.

B.Tech Courses

Core Courses

1. Discrete Mathematics and Graph Theory
2. Digital Circuits and Microprocessor
3. Data Structures and Program Design
4. Concepts in Programming Languages
5. Data Structures and Program Design
6. Computer Organization
7. Theory of Computation
8. Design and Analysis of Algorithms
9. Operating Systems
10. Software Engineering
11. Database Management Systems
12. Language Processors
13. Computer Networks
14. System and Network Security

Elective Courses

1. Introduction to Object Oriented Methodology
2. Neuro Fuzzy Techniques
3. Introduction to Web Programming
4. Software Engineering
5. Database Management Systems
6. Language Processors
7. Computer Networks
8. Steganography and Digital Watermarking
9. Advance Web Programming
10. Image and Video Processing
11. Introduction to Embedded Systems
12. Artificial Intelligence
13. Spatial Databases
14. Information Retrieval
15. Paradigms in Programming
16. Pattern Recognition
17. Advances in Compiler Construction
18. Formal Methods in Program Design
19. Introduction to Systems Engineering
20. Computer Vision
21. Introduction to Machine Learning
22. Data Mining &Data Warehousing
23. Introduction to Internet of the Things
24. Introduction to Distributed Systems
25. Topics in Graph Theory
25. Data Mining and Data Warehousing
26. Software Project Management
27. Business Intelligence
28. Introduction to Cloud Computing
29. Human Computer Interface
30. Software Architecture
31. Introduction to Parallel Computing
32. Distributed Systems
34. Number Theory and Modern Cryptography
35. Natural Language Processing
36. Deep Learning
37. Game Theory
38. Program Analysis
39. Advanced Computer Architecture
40. Topics in Bioinformatics

M.Tech Courses

Core Courses

1. Advances in Algorithms
2. Mathematical Foundation for Computer Science
3. Distributed System
4. Advanced DBMS
5. Cryptography & Information Security

Elective Courses

1. Pattern Recognition
2. Introduction to Systems Engineering
3. Soft Computing Techniques
4. Mathematics for Machine Learning
5. Deep Learning
6. Advances in Data Mining
7. Image Processing and Understanding

Student Activities

Apart from academic excellence, department of computer science and engineering encourages students to take part in various activities. Students from our department have participated, managed and led many inter departmental & inter college events. Our department secured first position Street Play, Stage Play, Institute gathering and won events like Smart India hackathon, Code for Good, Amazon's Hackon, Google's Code Jam. Such activities and events inculcates in our students the sound technical, managerial and teams skills needed to excel in life

Achievements

- A team emerged among the top 10 in the CodeStreet 21' by Amazon with more than 9000 participants.
- One student from our department was among the 30 girls selected from 9k+ participants at PAN India level for Twitter #DevelopHer program.
- One student was among 40 girls selected across India for the LinkedIn CoachIn mentorship program.
- One student is one of the 26 finalists from 2778 participants in the Instabase Encipher hackathon.
- 3 students managed to win the JPMC Code For Good - 2021 hackathon by contributing to their respective teams.
- One student secured Rank 2 in AxisHack, Hackathon by Microland.
  Two students from our department were part of a team whose research paper was accepted and published at the NeurIPS workshop, Pre-registration in ML.

- Dr. Shital Raut, Federated Learning For Estimating Air Quality, 2021 IEEE 12th International Conference on Computing Communication and Networking Technologies (ICCC).
- Dr. Syed Taqi Ali, "Traceable and Verifier-local revocable attribute-based signature with constant length" , International Conference on Security and Privacy (ICSP 2021) , NIT Jamshedpur,pp 132-146, 16-17 Nov. 2021.
- Dr. Ravindra B. Keskar, "Classification of Malignant Skin Cancer Lesion using CNN, KNN and SVM", International Conference on Data Science and Applications (ICDSA 2022), Jadavpur University, India. March 26-27, 2022.
- Total of 24 Journal papers, 11 conference papers were published and 2 books were authored.
- 2 students were Citi Campus Innovation Challenge Hackathon 4.0 Finalists

COLLECTIVE SKILLSET

B.TECH & M.TECH

- **Programming Languages:** C , C++ , Java , Python , JavaScript , Golang ,C# , Typescript
- **Web Technologies:** HTML, CSS , React , Angular , Node.js , Express.js , Django , Flask , Web Services , REST Framework , GraphQL
- **Databases:** MongoDB , MySQL , Oracle , Postgres , Microsft SQL, Redis
- **Cloud Technologies:** Docker , Kubernetes , Terraform , Azure , AWS ,
- **Frameworks/Libraries:** React , Angular , Node.js , Express.js ,Django , Flask , React Native , Flutter , Spring Boot Pandas ,Numpy, Requests , Beautiful Soup, Scrapy ,Matplotlib Seaborn, Scikit-learn , Keras , Pytourch , OpenCV ,TensorFlow , PyTorch

COLLECTIVE SKILLSET

B.TECH & M.TECH

- **Cyber Security :** IDS, IPS , Firewall management(PaloAlto) ,Watchguard , pfsense , opensense) , SIEMS (Azure XDR , RAPID 7 IDR) , IDAM ,AD Management(Azure AD , on prem AD) , Linux Hardening.

- **Virtualization :** VMware , Virtual Box , XCP-ng , Proxmox

- **Other Technologies:** Git , GitHub , Linux , Automation , UI/UX , Agile , Web Sockets , Networking , Splunk , Server Management (Windows , Linux) , Embedded Systems , IoT

- **Areas of Interest:** AI/ML , Data Science , Data Analytics , Deep Learning , Reinforcement Learning , Computer Vision , NLP , GenAI , LLM , Robotics , Blockchain , Full Stack Development ,Web Development,Backend Development , Database Management , Cloud Computing , Data Warehousing , Recommendation Engines , Pattern Recognitio , Data (ETL) Pipelines , Research , Cyber Security (Red Teaming) , Project Management , Cognitive Science , Human-Computer Interaction

PAST RECRUITERS

Google
amazon
APPDYNAMICS
D E Shaw & Co
citi
ORACLE
Goldman Sachs
VISA
WELLS FARGO
SAMSUNG
BNY MELLON
JPMorgan Chase & Co.
Morgan Stanley
airtel
HILTI
HDFC BANK
Un dos tres
accenture
TRILOGY INNOVATIONS
NCR
Capgemini
celigo
sling
magicbricks.com
Jio
ICICI Bank
Motion Reimagined.
Schneider Electric
Netcracker
incedo
DECiMAL POiNT Innovative Research Solutions
OLA ELECTRIC
PayU
cognizant
LOWE'S

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